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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 22.2

$$\begin{aligned}
 & \int \text{Bgsin}^2 x dx \\
 & \left\{ \begin{array}{l} u = \text{Bgsin}^2 x \\ dv = dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 2 \text{Bgsin} x \cdot \frac{1}{\sqrt{1-x^2}} dx \\ v = x \end{array} \right. \\
 & = x \text{Bgsin}^2 x - \int x \cdot \frac{2 \text{Bgsin} x}{\sqrt{1-x^2}} dx \\
 & = x \text{Bgsin}^2 x - 2 \int \frac{x \text{Bgsin} x}{\sqrt{1-x^2}} dx \\
 & t = \text{Bgsin} x \left(\Rightarrow x = \sin t, dx = \cos t dt, \sqrt{1-x^2} = \sqrt{1-\sin^2 t} = \sqrt{\cos^2 t} = \cos t \right) \\
 & = x \text{Bgsin}^2 x - 2 \int \frac{\sin t \cdot t}{\cos t} \cos t dt \\
 & = x \text{Bgsin}^2 x - 2 \int t \sin t dt \\
 & \left\{ \begin{array}{l} u = t \\ dv = \sin t dt \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = dt \\ v = -\cos t \end{array} \right. \\
 & = x \text{Bgsin}^2 x - 2 \left(-t \cos t + \int \cos t dt \right) \\
 & = x \text{Bgsin}^2 x - 2(-t \cos t + \sin t) \\
 & = x \text{Bgsin}^2 x + 2t \cos t - 2 \sin t \\
 & \text{Bgsin} x = t \\
 & = x \text{Bgsin}^2 x + 2 \text{Bgsin} x \sqrt{1-x^2} - 2x + C
 \end{aligned}$$

References